An Investigation into Predicting Vortex Ring State in Rotary Aircraft

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Abstract

Results from wind tunnel experiments and Computational Fluid Dynamics (CFD) simulations are presented which suggest it may be possible to predict when a rotary aircraft will enter Vortex Ring State (VRS). Results suggest that the formation of a small, toroidal vortex which develops approximately 2.5 blade radii above the rotor plane of a model helicopter develops over time as the helicopter descends into VRS. As the toroidal vortex develops, the lift the rotor is producing decreases over time until the toroidal vortex has reached the rotor plane of the helicopter. Repeating the experiment using the same model at different wind tunnel and rotor velocities produces the vortex state for particular ranges of values. Repeating the experiment using a propeller at two different but comparable propeller velocities shows a sudden increase in lift gradient for one of the velocities.

Nomenclature

a	Two-dimensional lift-curve slope
A	Rotor disc area
L	Lift
RD	Rate of Descent
V_i	Induced velocity
y^+	Non-dimensional wall distance
ρ	Air density
σ	Rotor solidity
θ_t	Blade tip angle-of-attack at hover

BET	Blade Element Theory
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
DES	Detached Eddy Simulation
LES	Large Eddy Simulation
MT	Momentum Theory
MP	Monitor Point
PIV	Particle Image Velocimetry
RANS	Reynolds-Averaged Navier-Stokes
RC	Remote Control
RPM	Revolutions Per Minute
UWE('s)	University of the West of England('s)
VRS	Vortex Ring State
VTM	Vortex Transport Model

Introduction

Recently there has been a renewed interest in understanding and predicting flow mechanisms which lead to a rotary aircraft entering Vortex Ring State (VRS). Both civil and military aircraft are expected to fly within a wide range of operating boundaries. If a rotary aircraft were to enter a powered near vertical descent, it is possible that the aircraft could enter VRS. If developed, VRS can cause a severe loss of lift and drastically increase pilot work load. This can and has led to fatalities in instances where both rotary and tilt rotor aircraft have entered VRS and crashed [1] [2]. When a rotary aircraft enters VRS, it will experience a dramatic loss in lift. This occurs as the rotary aircraft essentially descends into its own downwash produced from its rotor, typically through a reduction in airspeed with an increased rate of descent.

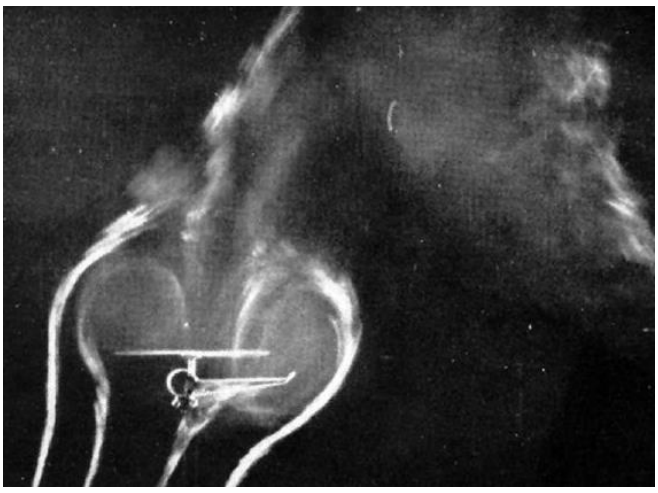


Figure 1.0: An illustration of a rotary aircraft in Vortex Ring State [3].

This causes vortices to circulate through the rotor disc of the aircraft and as a result, causes a severe loss of lift, as illustrated in figure 1.0. It should be noted that rotary aircraft

can enter VRS at any altitude; it is not linked to another aerodynamic effect in rotary aircraft called the ground effect. If a pilot enters VRS and tries to increase the rotational velocity in the rotor in an attempt to regain control of the aircraft, it would only intensify VRS and worsen the condition. The pilot needs to pitch the aircraft in a specific direction, so the rotor blades can 'cut' into clean, less turbulent air.

From pilots' experiences, three common VRS conditions have been derived. If these three conditions are met, a descent rate of 6000 feet per minute can develop. They are the following [4].

- A vertical or nearly vertical descent of at least 300 feet per minute.
- The rotor system must be using some of the available engine power (20-100%).
- Airspeed of 10 knots or less.

Some fixed wing aircraft have stall detection warning systems, to warn the pilot that he or she could be about to stall the aircraft. No such a system currently exists which is capable of warning in advance when a rotary aircraft will enter VRS.

The scope of work undertaken in this investigation is focused on renewing an understanding of how VRS develops and how a rotor transitions from a normal thrusting state to VRS. This involves understanding the nature of VRS and ways in which it may be possible to predict it. The majority of literature on helicopter aerodynamics mention or discuss VRS. It is fairly well understood in simplified terms how the flow regime changes from a normal thrusting state to VRS and what VRS looks like if you could see the air flow at the rotor plane. See figure 2.0 for a general illustration of how the flow regime changes as the descent rate of the helicopter increases (which illustrates the turbulent nature of VRS). The large inner vortex depicted in (b) will introduce a high level of turbulence into the rotor, increasing pilot work load and high vibration levels [5].

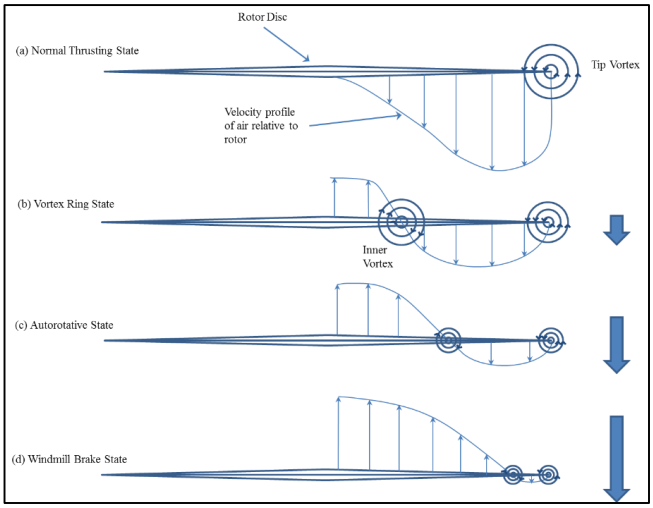


Figure 2.0: Induced Velocity Profile to Rotor Disc [6].

State (b) will cause a sudden and potentially unexpected loss of lift as unlike the autorotative state (c), VRS is a regime state that can be entered unexpectedly, whereas other thrusting states depicted in figure 2.0 are generally entered purposefully. This is what makes VRS a dangerous and potentially fatal operating condition. It should be noted that the frequency of such vibrations experienced when in VRS are not explicitly stated (other than simply stating that they are low frequency), leading one to believe there is room for a more precise predictive parameter relationship.

Helicopter flight tests highlight what conditions are required to cause a helicopter to enter VRS. By using well documented and understood helicopter aerodynamic theory (such as Momentum Theory (MT) and Blade Element Theory (BET)), conditions can be calculated which cause VRS to develop. Typically, VRS reaches its worst condition when the descent rate is three quarters the hover induced velocity [7]. Such forms of analysis can aid in understanding of VRS and define operating boundaries; however they cannot provide a detailed enough insight to create a VRS warning system. Mathematical models have been derived and developed to try and better understand the nature of VRS and in some cases try and predict the onset of VRS. Due to the highly unsteady nature of VRS, numerical approaches are often required to solve such models and visually display results for analysis. As stated by G. Ahlin & R. Brown [8]. "The unsteadiness in the rotor loading (during VRS) is caused by the collapse of the rotor wake into a highly time dependent, disordered flow with roughly toroidal topology, but the fluid dynamic mechanisms underlying this collapse remain a subject of active research". A common approach in mathematical modelling to try and better understand VRS is to model the rotor wake as it descends into its own downwash as it enters VRS. Such analysis has produced interesting results which have been expanded upon within this investigation's scope of work. For example, J. Bailly [9] demonstrated with an unsteady wake model (the MINT code developed by ONERA) that a sudden drop of local lift at the blade's tip and a high lifting area at the blade's root was observed as the rotor descended into VRS. Increases in fluctuations of rotor loads were also observed. In the authors' opinion, what is missing is a study to try and detect this sudden drop in local lift at the blade tip and high lifting area at the blade root. Advances in mathematical modelling have led to the development of operating boundaries for rotary aircraft. This has been achieved by again analysing the rotor wake under descent conditions to see at what critical point the wake collapses onto the rotor and forms VRS. However, conclusions from such studies [10] [11] have shown that quantification of the factors and conditions affecting the onset of VRS on rotors cannot be established based on descent velocity alone. This highlights the need for better quantification of VRS than just descent velocity and is further justified when VRS reportedly caused one of the modified Black Hawk helicopters to crash during operation Neptune Spear [12]. The Black Hawk helicopter was hovering above a compound

inside its walls, which were directing hot air towards the helicopter from higher than expected air temperatures. As the hot air rose towards the helicopters rotor it caused VRS to develop and subsequently the helicopter crashed. This highlights the need for another approach in understanding the development of VRS and a means to stop rotary aircraft unexpectedly entering VRS. Johnson el al. [13] have further developed momentum theory for calculation of the mean inflow in a rotor which is operating in VRS. Such work has been compared against flight test data and has shown good correlation. This has been used to further develop operating regions to avoid VRS and an algorithm has been developed to calculate the rotor induced velocity in VRS. This could in theory be used as a means to warn pilots they are about to enter (or are already in) VRS. However, the analysis is simplified due to momentum theory being used. Unsteady three dimensional flows are not being considered in the model, which would however be observed at the rotor plane in this instance. In the analysis that has been conducted in this area, it is rare to see several different approaches to understanding VRS being employed in the same study. Pure mathematical modelling approaches are used and are sometimes validated with third-party data; it is felt such studies could have benefited more if Computational Fluid Dynamics (CFD) were also used in comparison with flight and/or wind tunnel testing.

The purpose of wind tunnel testing in this investigations' scope of work is to obtain data on the model helicopter as it enters VRS along with the corresponding airflow at its rotor. Previous wind tunnel studies have been conducted using similar techniques. In some studies, propellers as opposed to an actual rotor have been used to create a wake and thus cause VRS to develop. Propeller force data and wake flow visualisation have been recorded as it enters VRS and in some instances as the propeller enters and leaves VRS [14] [15]. Such analyses have provided clues as to the nature of the vortex wake formation and shedding phenomena where flow visualisation dye is injected at the blades tip (but not in free stream flow). This particular experiment [14] was conducted in water as opposed to air, thus density, viscosity and other influential parameter values were different to a standard case and could affect flow conditions. Other experiments [16] state that the development of VRS is very sensitive to many of the rotors parameters; this would leave one to believe that data acquired from a propeller will be different to that acquired from an actual rotor.

Other wind tunnel experiments have been conducted that closer resemble what this investigation is planning to conduct. Such experiments do use 'off the shelf' Remote Control (RC) modelling equipment to operate dynamic models and measure forces acting upon the model as it enters VRS [17]. This study analysed the variations in rotor thrust at varying angles of attack and velocities when the rotors were operating in VRS. However, it does not highlight what happens to such parameters as the rotor descends into VRS over a period of time, something which this investigation aims to. Advanced and detailed wind tunnel experiments have been conducted which use detailed forms of flow visualisation such as Particle Image Velocimetry (PIV). PIV can obtain both flow visualisation and local flow data (such as velocity). Where this method of flow visualisation has been employed it has revealed complex flow within the vortex ring and wake flow (together with CFD analysis) and claimed that an expression for vortex ring and turbulent wake state onset has been developed [16]. The experiment used a scaled five bladed helicopter rotor. This investigation aims to put a model RC helicopter into VRS by mounting it vertically in the high speed section of The University of the West of England's (UWE's) wind tunnel on its measuring sting. Using helium bubbles, the airflow around the rotor will be visualised and photographed using a Panasonic HC-V10 video camera whilst the model helicopters lift is logged simultaneously. The use of helium bubbles to visualise airflow at the rotor disc as it enter VRS synchronised with measured lift data from the helicopter should be very beneficial for developing a means to understand the nature of VRS and how it develops (something which has not been attempted before as far as this literature survey suggests). The data extracted from wind tunnel testing will also be used to validate CFD simulations. This experiment will be repeated at different rotor speeds to better validate any findings.

Understanding flow mechanisms through and around rotors is a critical part of rotorcraft development and using CFD to analyse turbulent wakes is not just restricted to rotorcraft. Bodies which move through media fast enough (to cause flow separation and create turbulence) create turbulent wakes. Turbulence and wake modelling is still an active area of research where interest has been renewed in recent years with the advent of high performance computing and CFD.

Studies have been conducted that use CFD codes to analyse in detail the turbulent wake of a rotor as it descends into VRS. Because VRS is a very complex flow regime, studies have been conducted to see if both commercial and more specialist codes can model VRS. ElsA CFD solvers for Euler methods have been employed to try and understand how and why VRS occurs. Such studies [18] do look at velocity profiles on the rotor blade. But such studies are validated against MT, leading one to believe they could incur errors in doing so and fail to capture non-periodic flow. To combat this, more advanced codes and methods have been used. Using vorticity and Reynolds-Averaged Navier-Stokes (RANS) codes, blade root vorticity (something which has been highlighted in previously discussed work to be a sensitive parameter in the development of VRS) has been captured. However such methods do struggle to model diffusion and validation of codes is required [19]. Very detailed analysis using RANS/Detached Eddy Simulation (DES) and OVERFLOW codes has been performed with success to model rotor wake using a DES fluid model [20], but has not been conducted when the rotor is operating in VRS. Commercial codes such

as FLUENT have been proven to model wake and rotors operating in VRS [21]. The implementation of such codes modelled a helicopter descending into VRS to see how gaining forward airspeed mitigated the effect. However, this study assumed the rotor was a pressure disc, which caused a jump in pressure at its plane (actuator disc model). This would give a constant induced velocity (like MT) along the rotors radius, where in reality it would vary (as illustrated in figure 2.0). Despite this, the analysis does reflect common flow characteristics of VRS and was conducted with hopes to develop flight safety methods when in VRS – similar to this investigation. Other studies [22] have used well suited CFD codes to analyse a rotors wake when descending into VRS from both a hovering state and the windmill brake state (see figure 2.0). This particular study used Brown's Vortex Transport Model (VTM) to reveal the vortex dynamics that underpin the highly unsteady flow within VRS. The VTM uses an Eulerian grid based vorticity-velocity model when implemented within CFD to model rotor wake aerodynamics [23]. The VTM solves the unsteady vorticity transport equation which is obtained by taking the curl of the Navier-Stokes equation. This study did reveal in detail the vorticity field and its development within the toroid as the rotor descends into VRS and interestingly identified a zone of decoherence where clearly defined 'vortex worms' broke down into a high turbulent regions inside the toroid. This study did suggest that the collapse of the tubular wake produced from a rotor in a hovering state to VRS occurred in the absence of nonlinear feedback from the local velocity field to the strength of the wake. Other useful conclusions were made (some being qualitative) which agreed with previous studies. However, the simulation only used a single blade as a rotor to help simplify the problem being analysed which may in turn exclude other turbulent effects, this also meant any validation against raw data was not possible. Research has yet to show a study which looks in detail at a rotor operating in VRS where parameters such as velocity or pressure have been calculated as the rotor descends into VRS over time. Past analysis has covered aspects of this, but not in the level of detail hoping to be covered in this investigation with wind tunnel data that can validate such a simulation.

Reviewing literature from a variety of sources on three main topics relevant to this investigation does highlight 'gaps' in current knowledge. This investigation aims to fill such gaps and instead of looking to either further develop operating boundaries, aims to develop a concise means to detect when VRS will occur before it has. This is altogether a different approach in understanding the nature of VRS and how to avoid it, something which research has suggested aspects have been touched upon before but not expanded upon. It is hoped that even if this investigation is unsuccessful, wind tunnel data and other aspects of analysis will still be beneficial to others. As what is being attempted in this investigation has not previously been attempted before.

It has been theorised that distinctive fluctuations in pressure may be present at specific points on a rotorcraft rotor or propeller as it descends into VRS. If such fluctuations are present (or fluctuations of a different parameter) it may be possible to use these as means to design and implement a VRS warning system for rotary aircraft. To test this theory, a series of wind tunnel experiments have been devised which will measure the lift of a rotor as it descends into VRS. Helium bubbles will be used to visualise the flow of air around the rotor as it descends into VRS. CFD will be used to reaffirm any conclusions made from wind tunnel experimentation and to test if distinctive fluctuations in pressure are present at any point on the rotor.

Wind tunnel Experimentation

Literature states that VRS should reach its worse condition when the rate of descent is three quarters the induced velocity in hover [7]. It is to be assumed that the model helicopter is operating at full power when being tested in the wind tunnel, as some of the available power (20% or more) needs to be used in order for a rotary aircraft to enter VRS. One would therefore assume that by using the maximum available power would create the most influential VRS condition the rotary aircraft could experience. Using MT and BET to calculate the mean induced velocity yields the following results.

Parameter	MT	BET
Mean Hover Induced Velocity	2.88 m/s	2.91 m/s
Wind Tunnel Velocity	2.15 m/s	2.18 m/s
Angle of Attack at Hover	-	5.4 °

Table 1.0: Summarised Theoretical Test Conditions

The model helicopter used in wind tunnel experiments is an Align T-Rex 250 which has been mounted onto the measuring sting in UWE's subsonic wind tunnel, allowing for the lift force its rotor produces to be measured over time. Upstream of the model helicopter is a nozzle mounted on a stand through which helium bubbles are injected into the airflow which will pass through the model helicopters rotor. Downstream of the model helicopter is a high intensity lamp which illuminates the helium bubbles allowing for flow visualisation of the airflow as it passes through the rotor. A Panasonic HC-V10 video camera is mounted on a tripod perpendicular to the rotor photographing the experiment. Refer to figure 3.0 for an image detailing the wind tunnel experiment. Keeping the wind tunnel velocity at a constant 2.2 m/s with the model helicopter in position, the video camera started photographing the experiment and lift data was logged over a range of sample frequencies (7.7 Hz, 20 Hz and 33.3 Hz) (Stage 1 of the experiment). Three seconds later the model helicopter started to accelerate towards its required test conditions which it reached approximately fourteen seconds into the experiment (Stage 2 of the experiment). Twenty-five seconds into the experiment the model helicopter was stopped (Stage 3 of the experiment) and after thirty seconds from the start of the experiment both

the video camera stopped photographing the experiment and logging of lift data was stopped (Stage 4 of the experiment).

Due to the highly turbulent, three dimensional and time dependent nature of VRS, the data sets for their respective sampling frequencies have been averaged (15 samples for 7.7 Hz, 30 samples for 20 Hz and 20 sampled for 33 Hz) and are present in figure 7.0.

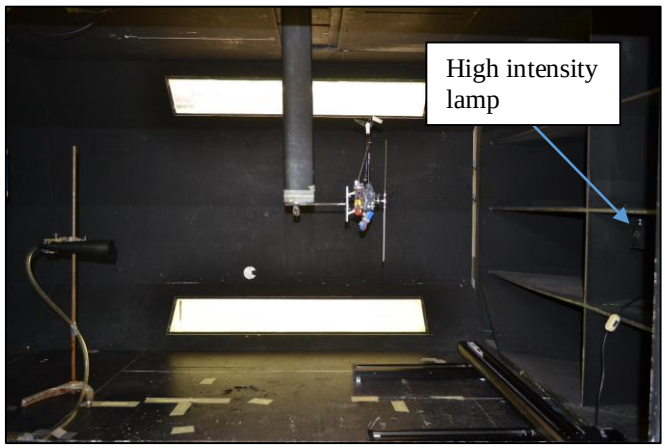


Figure 3.0: Wind Tunnel Experiment

Note that lift is plotted as the model helicopter was mounted vertically, where its rotor produces lift (which is measured as drag through the wind tunnel). When analysing figure 7.0 there is an apparent change in continuity which occurs between approximately 8 and 9.5 seconds in the plot, where lift starts to decrease before beginning to increase again. Interestingly, this change in continuity happens at the same time a vortex is formed and reaches the rotor plane of the model helicopter. This development of the vortex over time is present in figure 5.0. Fluctuations are clearly present in figure 7.0 during stage 3, where the model helicopter is operating in VRS. Literature documents that low frequency vibrations are possible when a rotor is operating in VRS [5] [13], as previously stated such literature does not state the frequency of such potential vibrations. By analysing the peaks present in stage three of figure 7.0, peaks are spaced approximately 0.5 seconds apart. Therefore the frequency of vibrations experienced during VRS is equal to 2 Hz. This apparent change in continuity may also constitute a distinctive pressure fluctuation at a point on the rotor blades, as the vortex is generated and moves closer to the rotor plane. Figure 6.0 shows reflow through the rotor when the vortex reaches the rotor plane, this reflow occurs at approximately half the blades radius. This would lead one to believe that between one quarter and three quarters the blades radius a distinctive pressure fluctuation could be observed from these respective points on the blade. The data extracted from wind tunnel experiments together with flow visualisation data that has been synchronised with wind tunnel data should serve as a means with which to validate a CFD simulation with.

When analysing the results from wind tunnel testing present in figure 7.0, it was theorised that the change in continuity present at approximately eight seconds could be attributed to the formation and growth of the vortex shown in figure 5.0 taking kinetic energy from the rotor. This theory can be tested by measuring the Revolutions Per Minute (RPM) of the model helicopters rotor as it enters VRS and synchronising it with respective lift data from the wind tunnel. As rotational kinetic energy is a function of the moment of inertia (a constant) and the rotational velocity squared, a similar change in continuity of angular velocity should be present at approximately eight seconds when plotted against time. An Eagle Tree V4 data logger with optical infra-red RPM logger was installed on the T-Rex 250 model helicopter where it could accurately log the RPM of the main gear (which rotates at the same rotational velocity as the main rotor). The Eagle Tree V4 data logger was then programmed to log RPM at 40 Hz. The wind tunnel experiment was then repeated 20 times with lift produced by the rotor logged at 33 Hz. A further 20 experiments were performed where the wind tunnel was not operating, this was conducted to test that the change in continuity observed at eight seconds in figure 7.0 can be attributed to the rotor descending into VRS when the formation of the vortex occurs in figure 5.0. This extended experiment has produced the following data, present in figure 8.0. When analysing the data present in figure 8.0, no apparent change in the continuity of rotational velocity is present at eight seconds when compared to the change in continuity of lift at the same time when the rotor is descending into VRS. Interestingly, the change in continuity of lift is not present when the wind tunnel is not operating, thereby not putting the rotor into VRS. This is strong evidence that the change in the continuity of lift present at approximately eight seconds in figures 7.0 and 8.0 can be attributed to the rotor entering VRS and the formation of the vortex in figure 5.0.

To better understand the nature and limits of this discontinuity in lift and vortex formation, a series of subsequent wind tunnel experiments were performed in the same manner as have been previously outlined and discussed, but at a combination of different induced and descent velocities, as shown in figure 4.0.

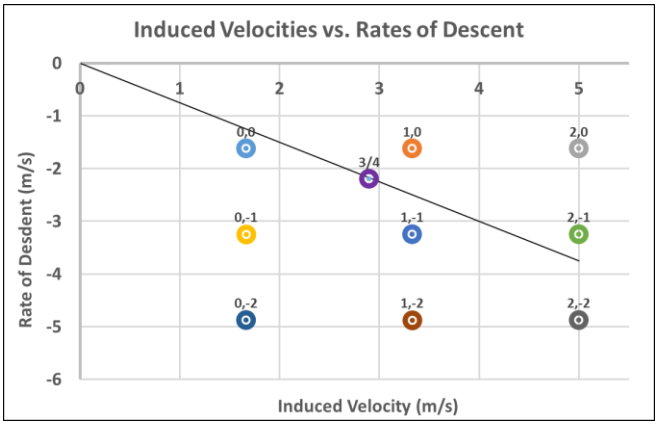


Figure 4.0: Different Induced Velocities and Rates of Descent

Results from testing at conditions illustrated in figure 4.0 are presented in figure 9.0 and 9.1. Results from the nine new test conditions do show that the discontinuty in lift experienced at approximatley eight seconds is somewhat sensitive to the ratio of induced velocity to rate of descent. Data series, “1,0”, “2,0” “1,-1” and “2,-1” all experience a discontinuty in lift (some not being as severe as identified in figure 7.0 or “3/4”). Each data set presented in figure 9.0 were averaged out over five repeated tests, where data was logged at 20 Hz. Figure 9.2 illustrates this effected region by overlaying an estimated rounded rectangle over the effected conditions. Lift has been non-dimensionalised in this presented data to create a regieme diagram by dividing it by the mass flow rate through the rotor, multiplied by the rate of descent, as shown in equation 1.0.

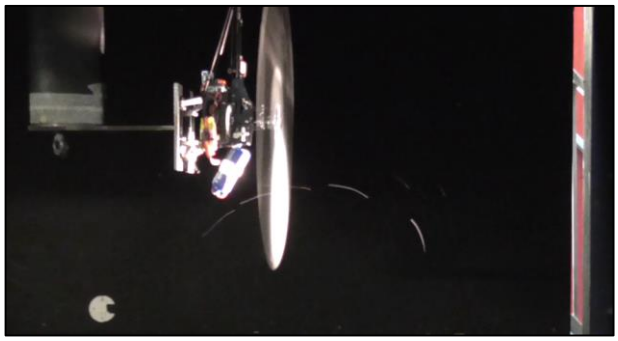


Figure 6.0: Recirculation Vortex through Rotor Plane

$$\frac{L}{\rho \cdot A \cdot V_i \cdot RD} \qquad (1.0)$$

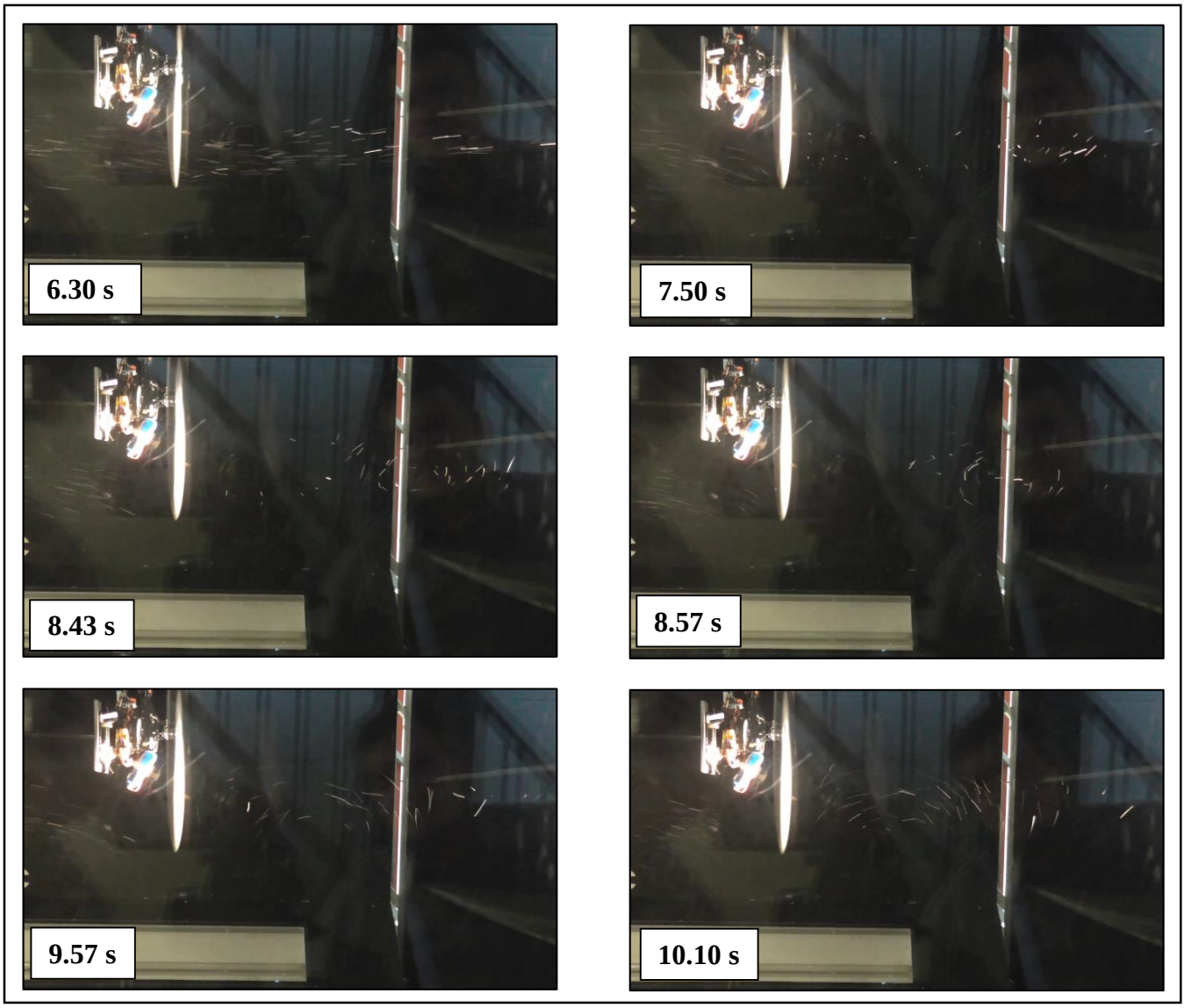


Figure 5.0: Vortex Development

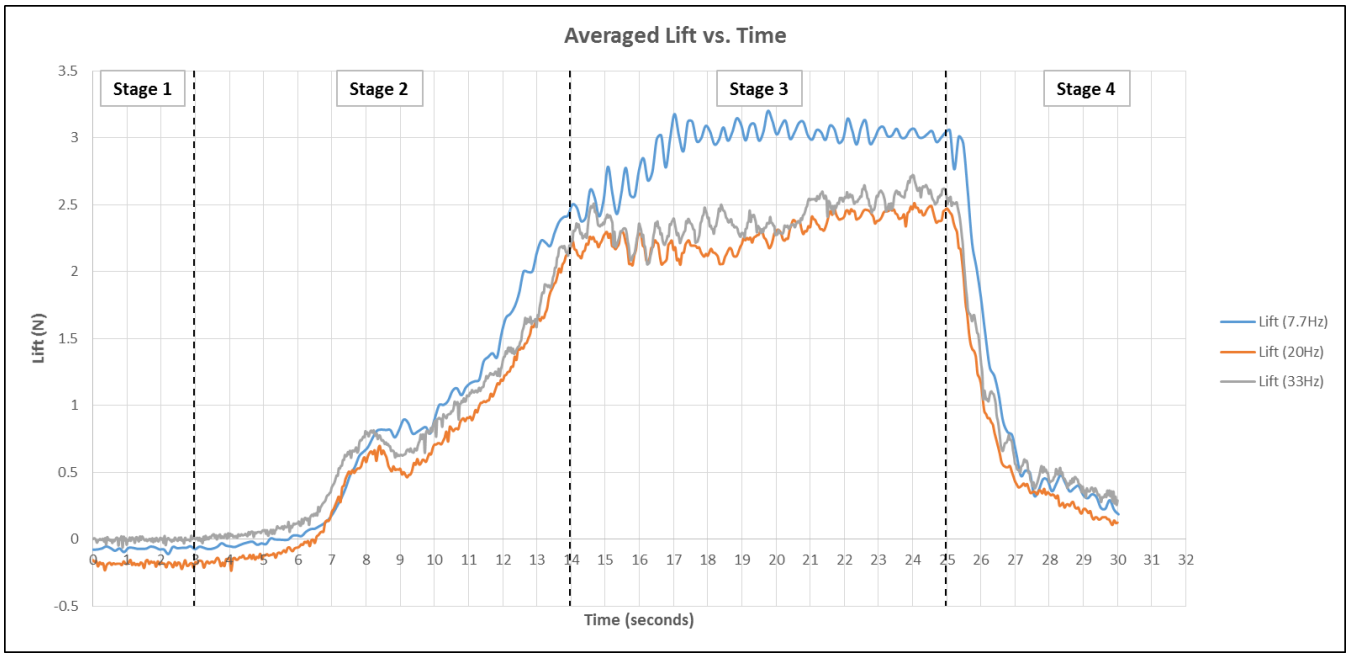


Figure 7.0: Averaged Lift vs. Time for 7.7 Hz, 20 Hz and 33 Hz Data

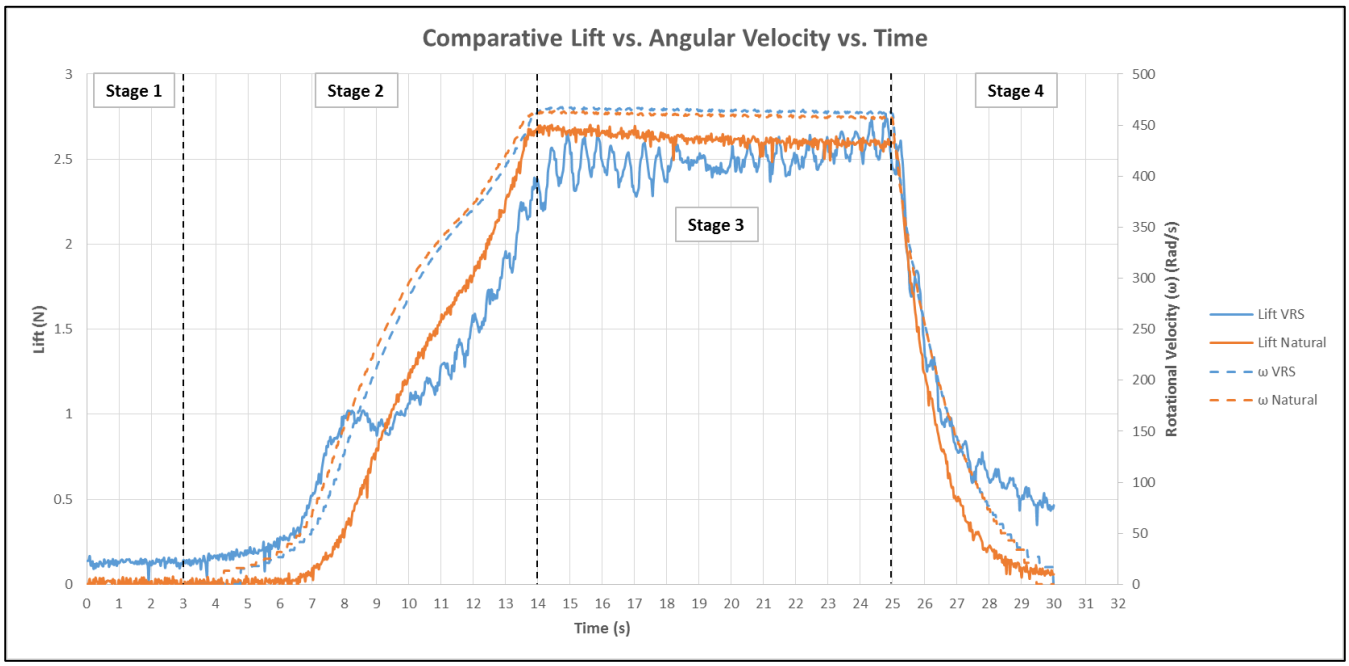


Figure 8.0: Averaged Lift and Rotational Velocity vs. Time

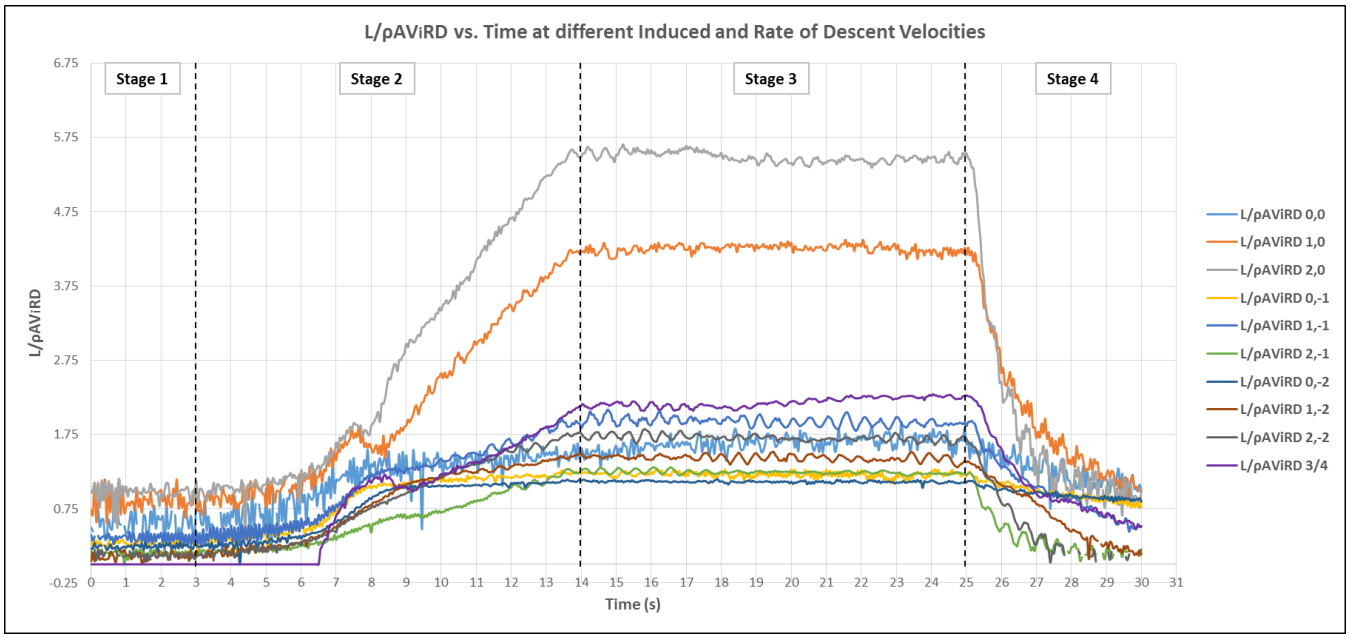


Figure 9.0: L/ρAViRD vs. Time at varying Induced and Descent Velocities

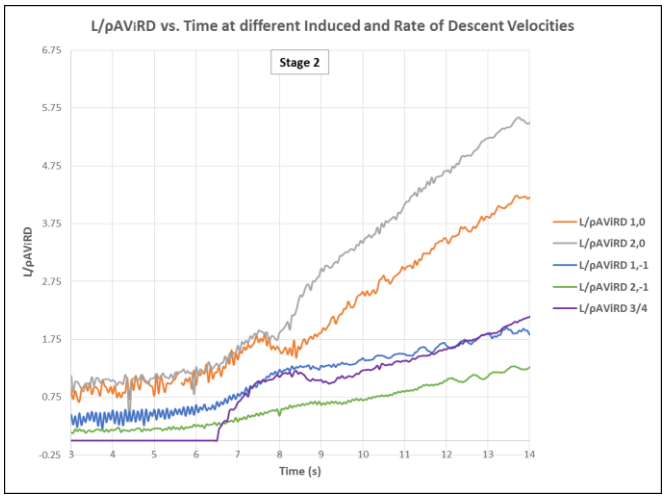


Figure 9.1: L/ρAViRD vs. Time at varying Induced and Descent Velocities – Effected Conditions

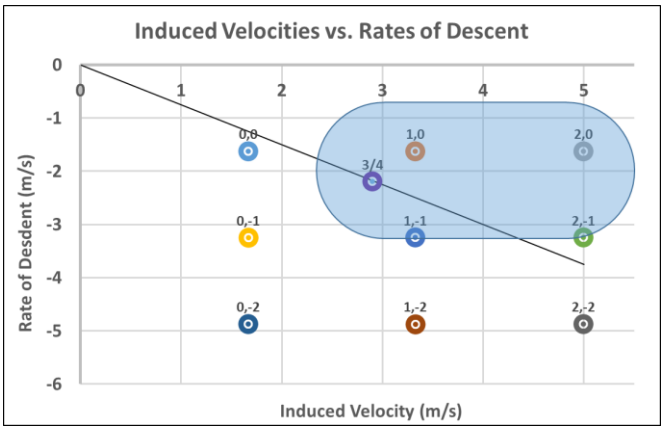


Figure 9.2: Different Induced Velocities and Rates of Descent – Effected Conditions

All results presented and discussed so far have used the same model helicopter, thus the geometry in all results is constant. Such results could just be specific to the T-Rex 250 model helicopter when operating in the conditions presented, therefore, in an attempt to eliminate this possibility, a propeller has been tested in UWE's subsonic wind tunnel using the same speed controller and lithium polymer battery as was used when testing the model helicopter. This new experiment is shown in figure 10.0.

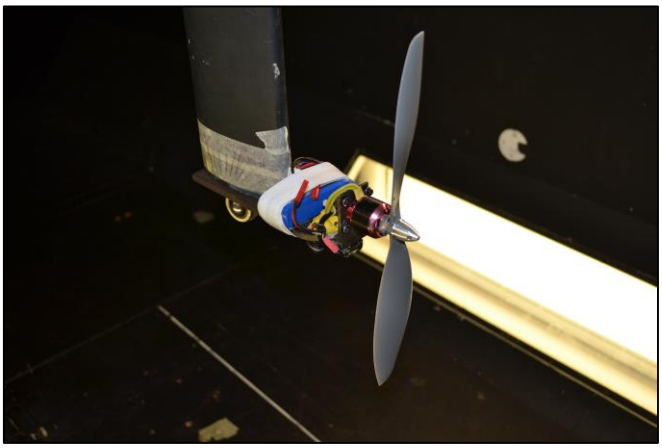


Figure 10.0: Propeller Experiment

The propeller is 279.4 mm in diameter and has a pitch of 119.38 mm, the motor is a 16 amp Extreme Flight 22T/930. To closely replicate the previous experiments with the model helicopter, the mean induced velocity was set at 2.9 m/s with a wind tunnel velocity of 2.2 m/s (three quarters the mean induced velocity). This mean induced velocity was produced at 23.5% throttle (not 100% throttle in experiments using the model helicopter), this means the acceleration of the propeller in this experiment is not identical to acceleration of the model helicopter rotor, which took eleven seconds to reach 100% throttle as opposed to five seconds using the propeller. This experiment was repeated twenty times with lift data being logged at 33.3 Hz, results from this experiment can be found in figure 13.0.

To test the sensitivity of propeller acceleration, the experiment was repeated a further twenty times with lift being logged at 33.3 Hz but with the propeller running at 100% throttle. This means acceleration is now comparable to the model helicopters rotor acceleration, the mean induced velocity was measured at 7.91 m/s when operating at 100% throttle, therefore the wind tunnel velocity was set and three quarters this velocity (5.9 m/s). Results from this experiment can be found in figure 14.0.

When analysing the data in both figures 13.0 and 14.0, it is clear that the acceleration of the propeller is a very influential factor in the nature of its descent into VRS. Figure 13.0 does not appear to show any clear discontinuity in lift when compared to figure 7.0. However, figure 14.0 appears to show a sudden change in gradient at approximately 6.5 seconds, such a sudden change could potentially be used as a means to trigger a VRS warning system, but could be attributed to the natural acceleration of the rotor. Low frequency vibrations are also observed in figure 14.0 when operating in VRS of approximately 2 Hz.

Computational Fluid Dynamic Analysis

CFD simulations have been conducted to test if a distinctive fluctuation in pressure is present at a point on the model helicopters blade(s) and if the vortex development shown in figure 5.0 can be recreated within a numerical simulation. ANSYS 14.5 CFX software has been used to test this. The wind tunnel's geometry has been replicated in Computer Aided Design (CAD) software SolidWorks, with a simplified model of the T-Rex 250's rotor, shown in figure 11.0.

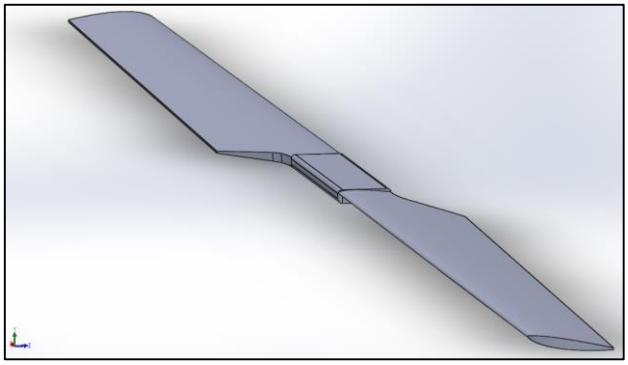


Figure 11.0: Simplified T-Rex 250 Rotor

An additional CAD model of a sphere which encapsulates the rotor has been created which has a much greater mesh density than the wind tunnel. This should better capture the vortex formation above the rotor plane and increase fidelity in the pressure experienced at the rotor, where Monitor Points (MP) shall be placed to log both pressure as the rotor descends into VRS and vorticity above the rotor in three locations to detect the development of the vortex in figure 5.0.

As the formation of the vortex illustrated in figure 5.0 occurs approximately 2.5 blade radii downstream from the centre of the rotor (≈ 573 mm), the inflation sphere has a radius of 600 mm. Figure 12.0 shows these three domains.

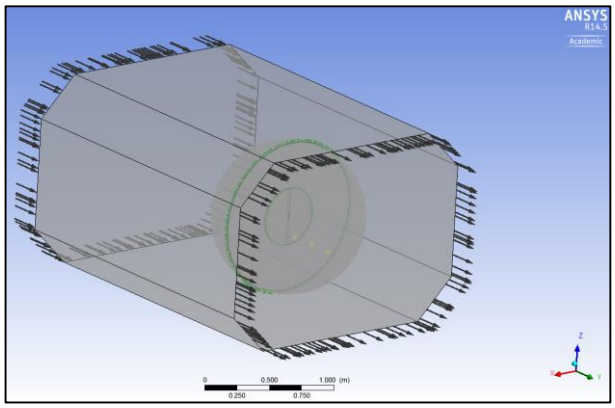


Figure 12.0: Simulation Domains and Conditions

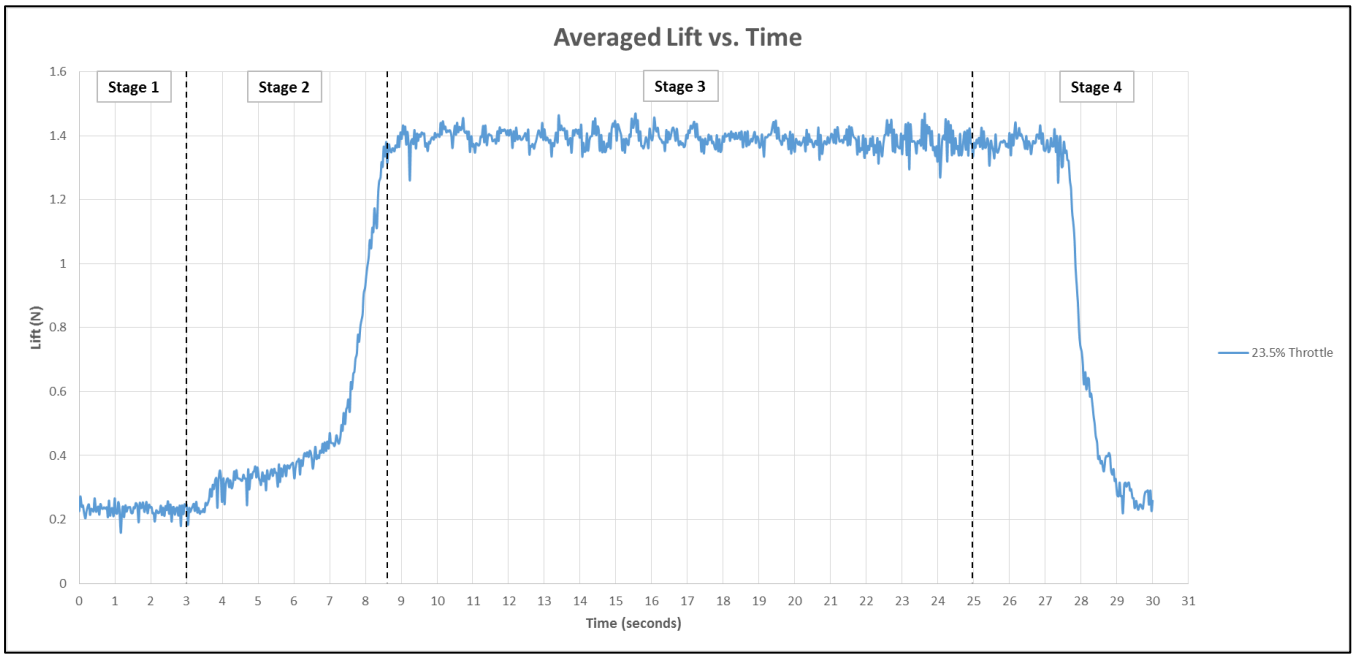


Figure 13.0: Averaged Propeller Lift vs. Time

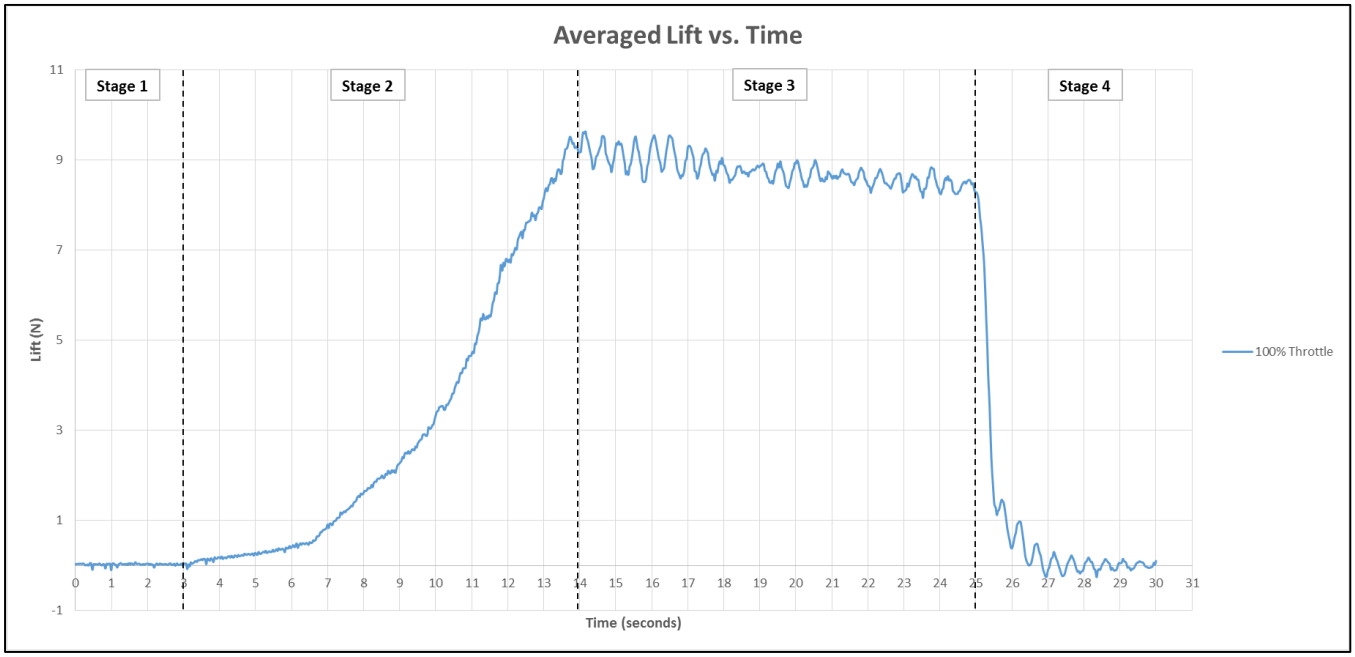


Figure 14.0: Averaged Propeller Lift vs. Time

A preliminary simulation has been created which uses a RANS turbulence model (K-Epsilon). The preliminary simulation is transient and has a time step of 0.05 seconds (20 Hz); this allows for a more direct comparison between the simulation and the wind tunnel data which was logged at 20 Hz. If this preliminary simulation does not correlate well with wind tunnel data or is unsuccessful in modelling the vortex development shown in figure 5.0, a second simulation will be created with modification to initial conditions. The mesh for the preliminary simulation has been created using ANSYS meshing software to create an unstructured mesh based on Large Eddy Simulation (LES) meshing requirements with a y+ value of 100 [24]. Initial conditions of the simulation replicate those experienced during wind tunnel experiments. Results from the preliminary CFD simulation show good correlation with wind tunnel data when comparing the lift produced by the rotor (figure 15.0); however the preliminary simulation does not appear to capture the formation of the vortex shown in figure 5.0 and subsequent discontinuity in lift. The preliminary CFD simulation did suggest that distinctive fluctuations in pressure were present at the tip of the rotor when descending into VRS. These fluctuations manifested themselves as short, momentary increases in pressure as shown in figure 16.0. Subsequent three dimensional vortex core mapping of the airflow revealed that all of these momentary increases in pressure were due to a small, localised vortex coming into contact with the tip of the rotor, giving it a short but sharp momentary increase in pressure, as shown in figure 19.0.

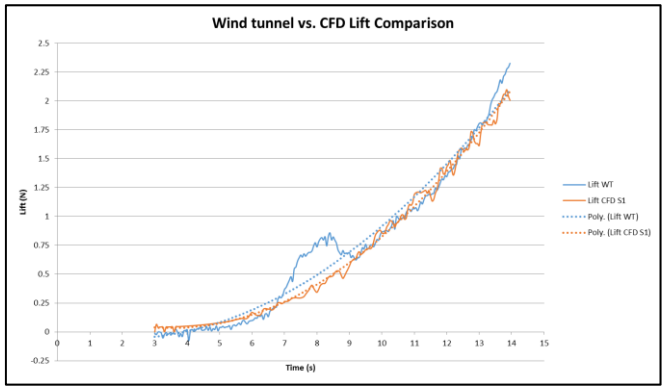


Figure 15.0: Wind tunnel vs. CFD Data

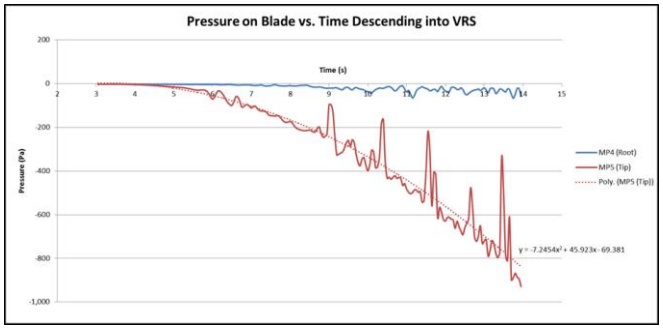


Figure 16.0: Pressure Fluctuations on Rotor Blade

The preliminary CFD simulation was modified to try and capture the vortex formation shown in figure 5.0 by having common frame change models between the rotor and inflation sphere domains (frozen rotor), as this may better communicate flow information between them. This modified simulation was also computed using high resolution numerics, but failed to capture the vortex formation in figure 5.0 (see figure 17.0). However, these two modifications changed flow properties to the extent that fluctuations in pressure were no longer present at the tip of the rotor, as shown in figure 18.0.

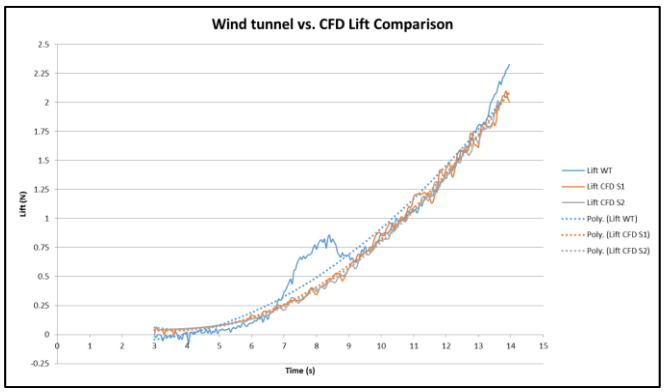


Figure 17.0: Wind tunnel vs. CFD S1 and S2 Lift Data

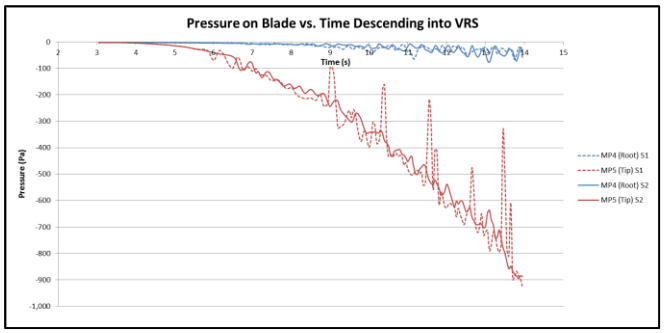


Figure 18.0: Pressure Fluctuations on Rotor Blade

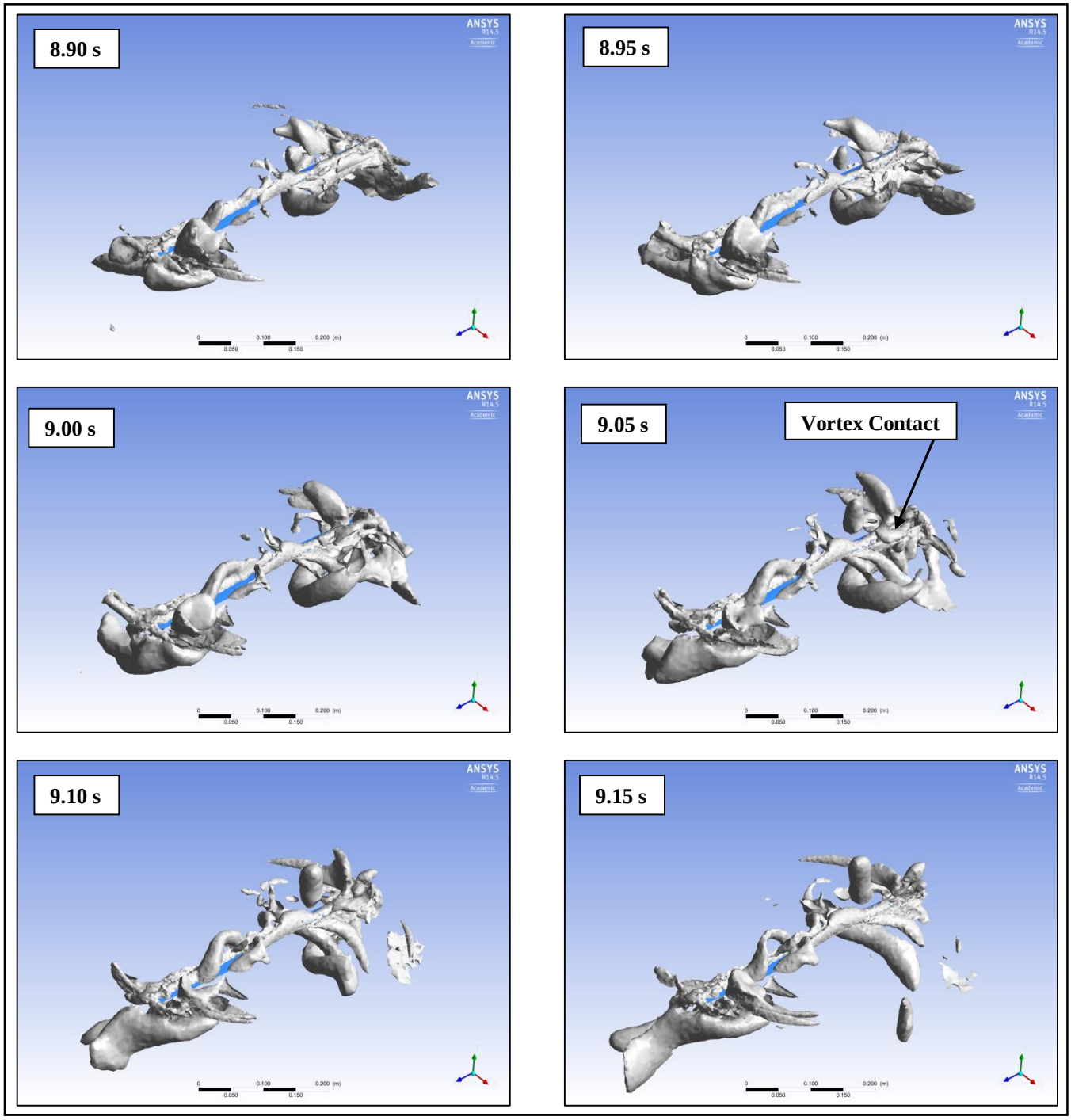


Figure 19.0: Vortex Core Region Development

Conclusions

This investigation has shown that it could be possible to predict when a model helicopter will enter VRS by measuring its lift in real time and using the discontinuity in lift detected during wind tunnel experiments as a means to 'trigger' a VRS warning system. Low frequency vibrations of 2 Hz were measured during wind tunnel experiments when the rotor was operating in VRS (figure 7.0).

Further wind tunnel experiments at a combination of different induced and descent velocities and the use of a propeller as opposed to a model helicopter's rotor has highlighted boundaries in these velocities and influential parameters, which appear to create a distinctive fluctuation in lift for test some cases. The rate of descent into VRS (the acceleration of the rotor/propeller) appears to have a strong influence in the nature of descent into VRS and if such fluctuations are observed.

CFD simulations were in good agreement with wind tunnel data; however the simulation failed to capture the formation of the vortex shown in figure 5.0. This could be because of the nature of the RANS turbulence model averaging out turbulence over time, thus stopping the vortex from forming and developing over time (as shown in figure 5.0).

Initial CFD simulations also suggested that fluctuations in pressure were present at the tip of the rotor (figure 16.0). However, once numerical modifications/improvements were made to the simulation, these fluctuations were no longer present (figure 18.0). This would lead one to believe that the simulation is not yet accurately modelling the turbulent, three dimensional flow of air around the rotor as it descends into VRS.

Recommendations for Future Work

If future work is conducted which follows the work presented in this paper, it is recommended that a larger wind tunnel section be used when conducting wind tunnel experiments to ensure wall effects are negligible. The use of a different model with different geometry and with a variety of descent rates into VRS (rotor/propeller acceleration) would also help further validate the findings made in this investigation. As this investigation has shown that the rate of descent into VRS has a strong influence on the fluctuation in lift being observed, if it could be proven that such fluctuations are still present within actual rotorcraft operating boundaries then this could be used as a means to design and implement a VRS warning system. Better synchronisation methods and flow visualisation methods could be employed in such a study.

Future CFD simulations should use a more advanced turbulence model (scale resolving simulation or better). As although the RANS simulation was in good agreement with wind tunnel data, it did not capture the formation of the vortex shown in figure 5.0 and subsequent discontinuity in lift. If a turbulence model was employed in such a simulation which did not average turbulence over time it may capture the vortex formation and subsequent discontinuity in lift. Structured meshing (as opposed to unstructured meshing used in this investigation) would also improve such a subsequent simulation and should be considered. The use of a moving mesh to model the rotation of the rotor as opposed to a rotating domain may produce different results again so should also be considered in any future work.

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Align T-Rex 250 Specifications

Main rotor diameter [25]	460 mm
Tail rotor diameter [26]	108 mm
Length [26]	430 mm
Height [25]	150 mm
Width [26]	86 mm
Total mass (measured)	342 grams
Main rotor RPM (calculated)	4481.625
Main rotor RPM (measured)	4430
σ	0.07322
a	5.14
θ_t	5.4°/0.09425 Rad
A	0.165 m^2